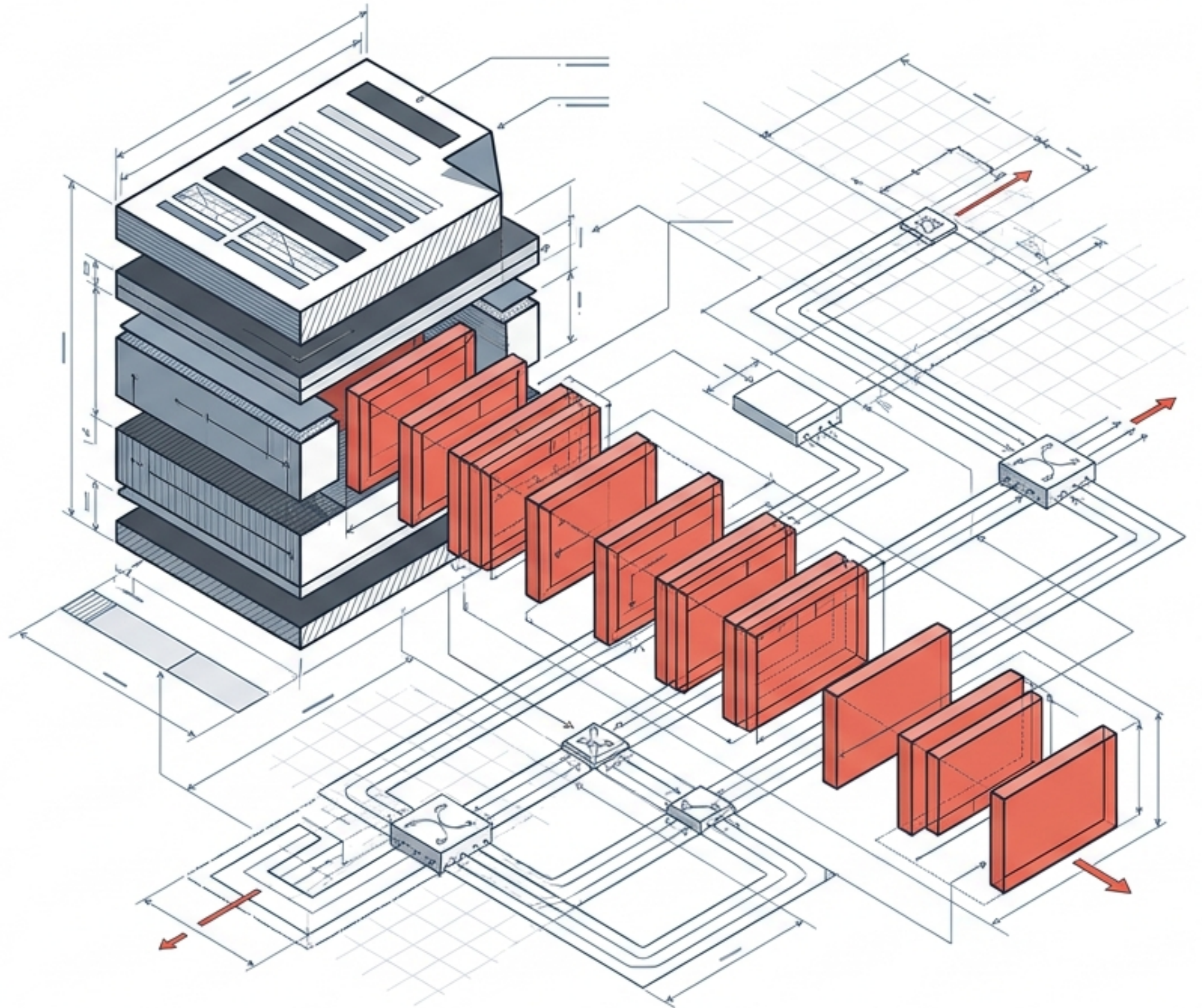


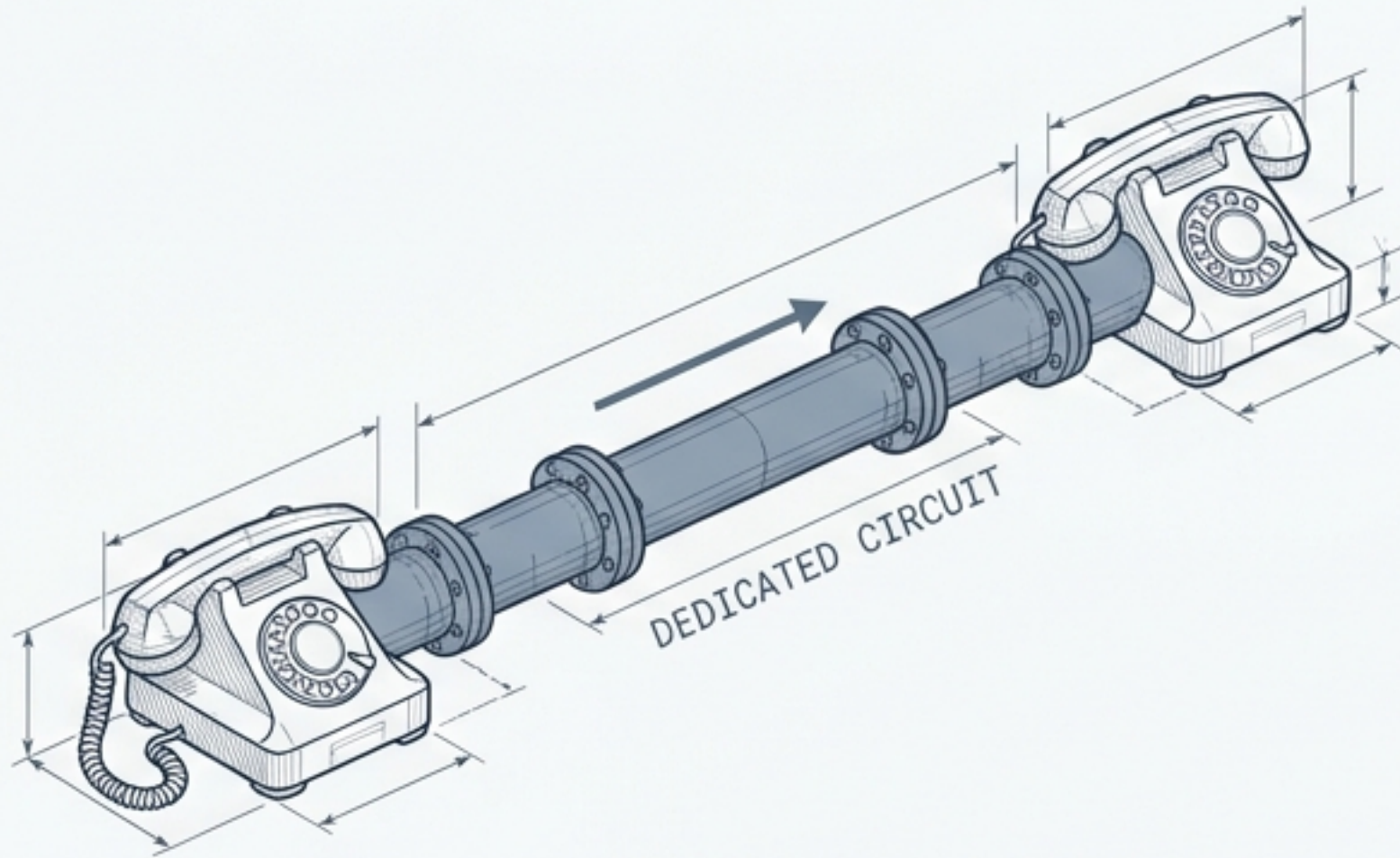
The Anatomy of Data Flow

Packets, Frames, and How Information Travels the Internet

A visual guide to networking paradigms, **encapsulation**, and **performance**.

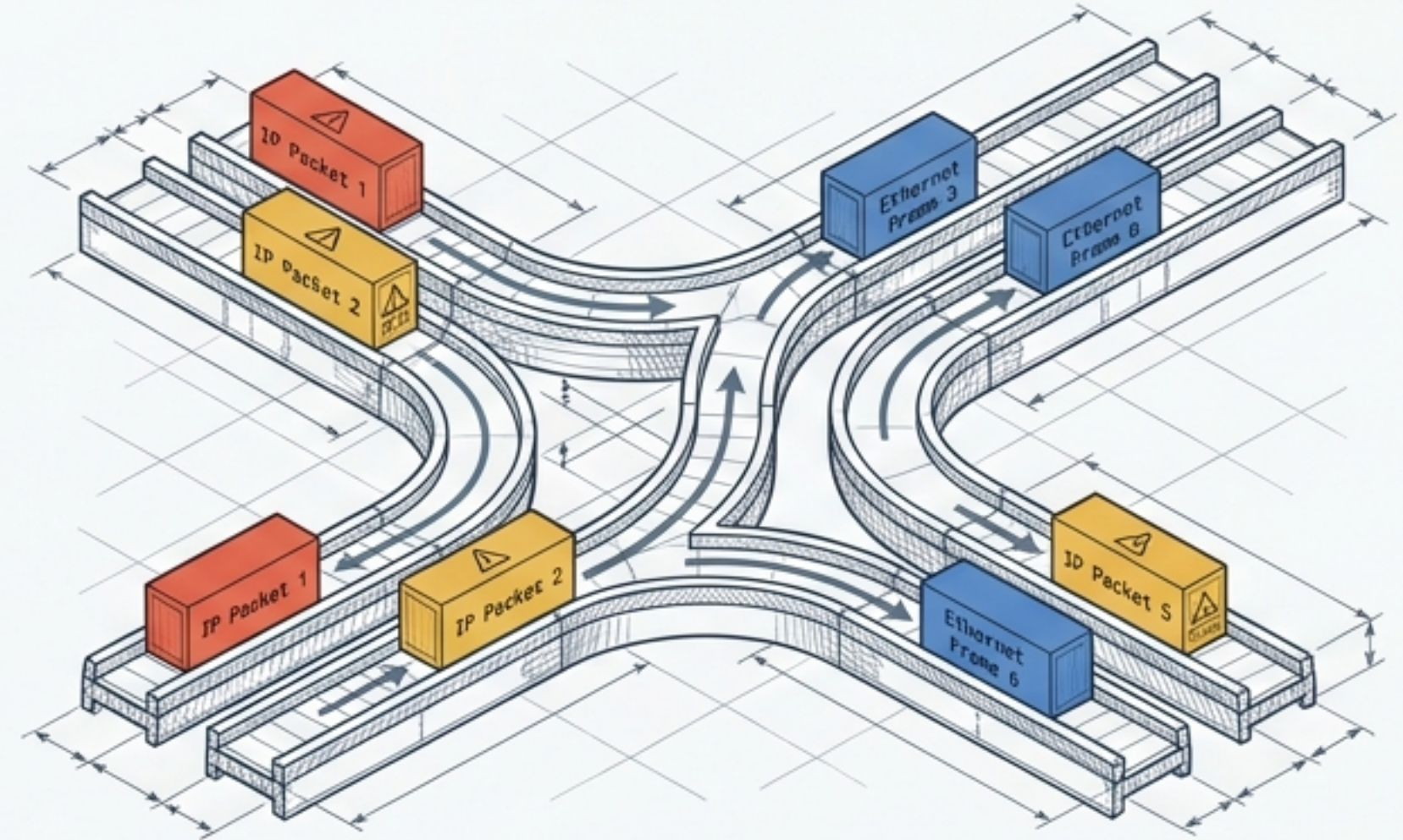


Circuit Switching vs. Packet Switching



Circuit Switching

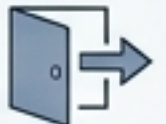
- Establishes a dedicated, reserved communication path before transmission.
- Highly predictable, but highly inefficient for modern bursty data.
- Resources are locked and wasted even when completely idle.



Packet Switching

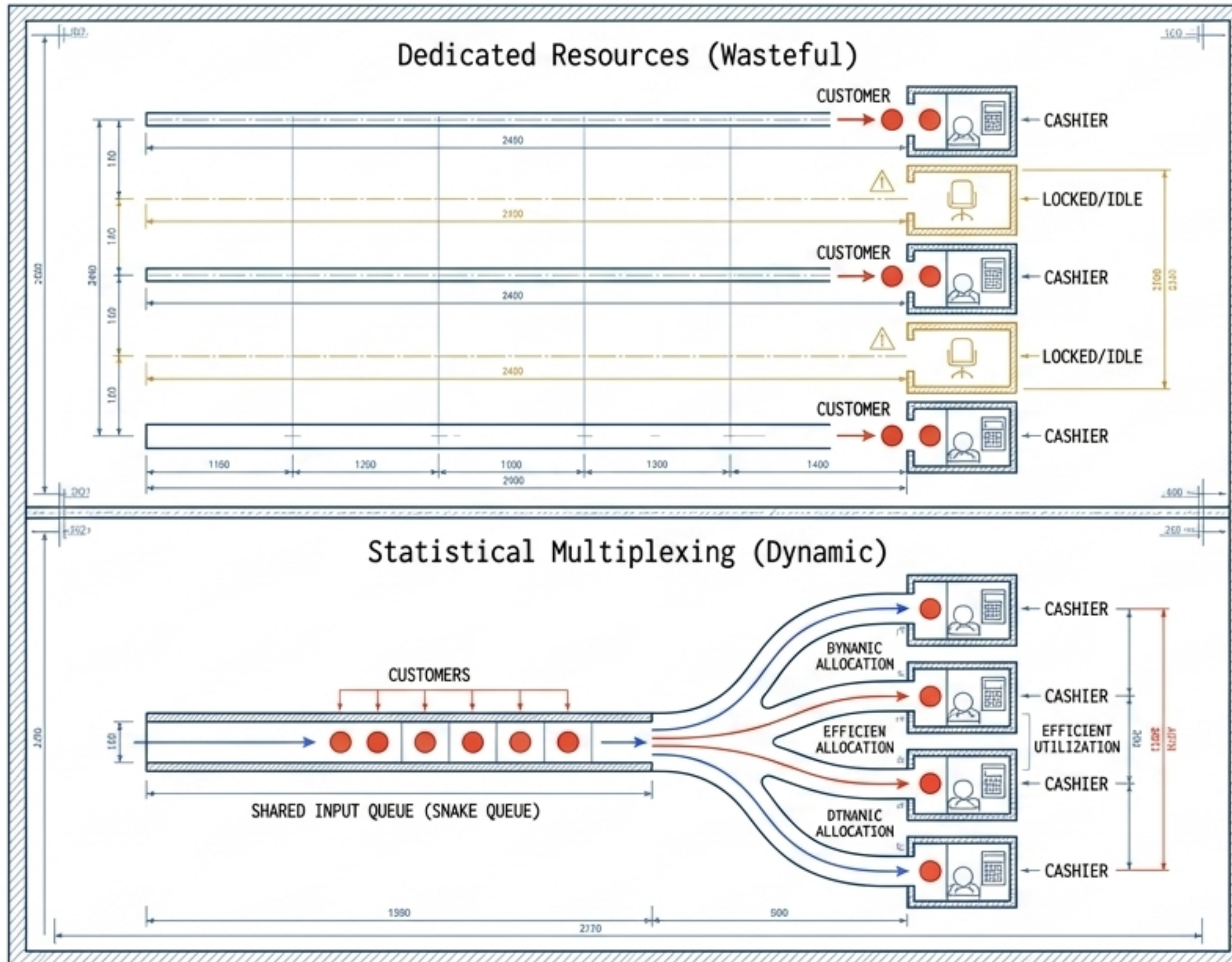
- Data is divided into small, independent units.
- No permanent path exists between devices.
- The network is dynamically shared, transmitting data only when capacity is available.

Why the Internet Chose Packet Switching

Feature	Circuit Switching	Packet Switching
Communication Path	 Dedicated	 Shared
Resource Reservation	 Required	 Not Required
Bandwidth Utilization	 Fixed & Wasted	 Dynamic & Efficient
Scalability	 Severely Limited	 Excellent
Fault Tolerance	 Lower	 Higher

The Scaling Reality: If the Internet used circuit switching, the sheer volume of unpredictable, bursty web traffic would require vastly more physical infrastructure than exists today. Only devices actively transmits consume bandwidth in a packet-switched network.

The Secret to Scale: Statistical Multiplexing



■ The Unpredictability of Traffic:

- Users constantly alternate between clicking, downloading, and remaining idle.

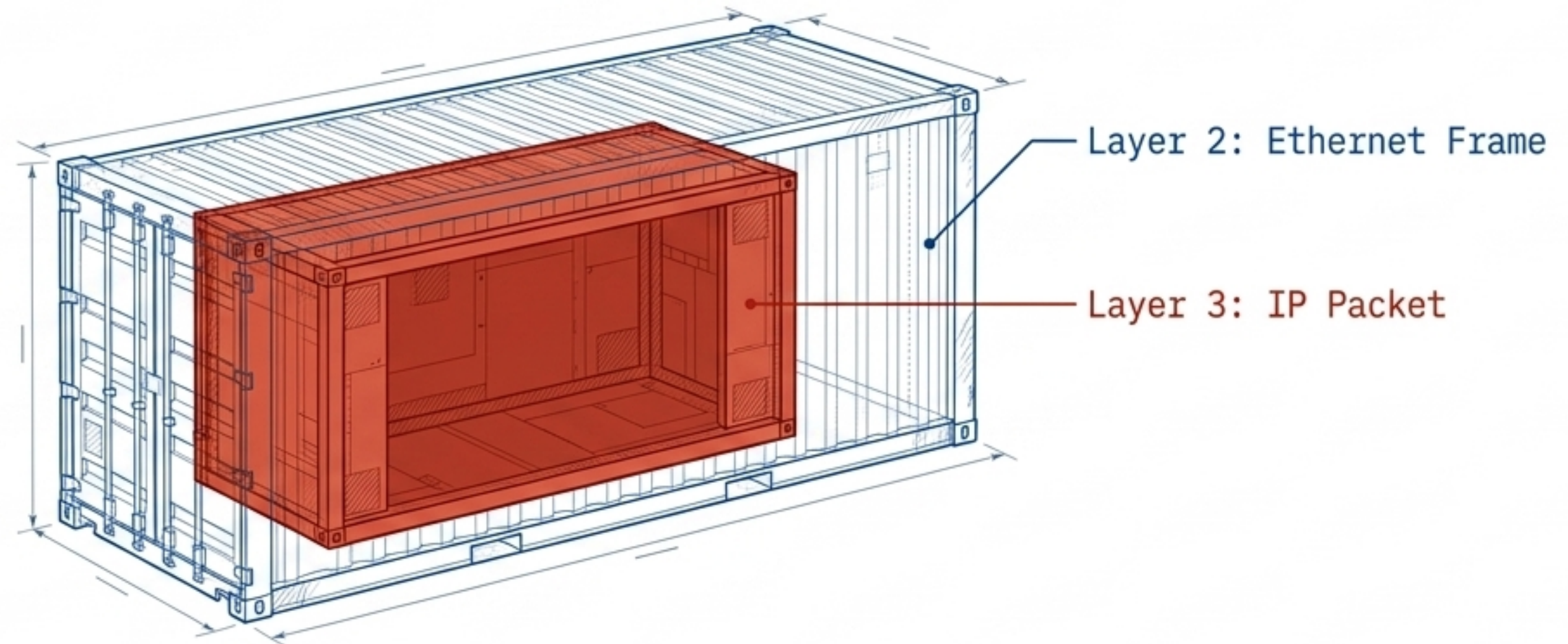
■ The Mechanism:

- Instead of permanently reserving bandwidth, the network dynamically shares available capacity among active users.

■ The Result:

- Whenever one user becomes inactive, another's packets instantly utilize the freed bandwidth. This allows massive scale with limited physical resources.

Packets vs. Frames: Not the Same Thing



The Frame (Link Layer)

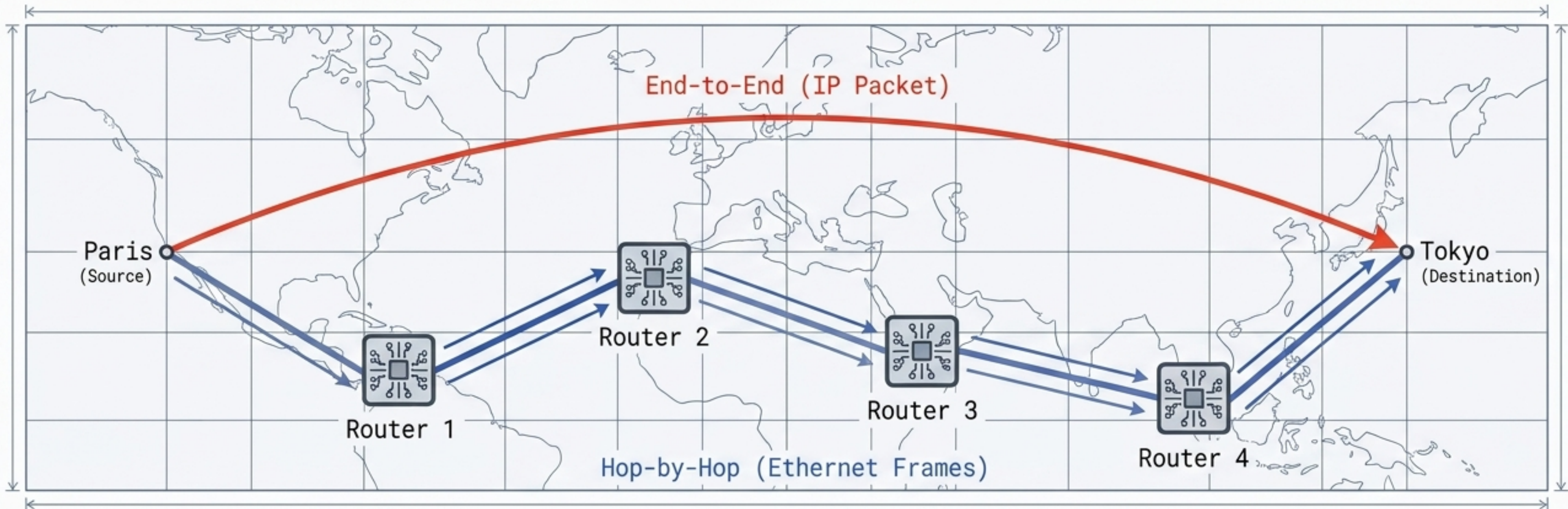
- The temporary outer container.
- Exists exclusively for communication across a single physical link.
- Destroyed and rebuilt at every router.

The Packet (Network Layer)

- The inner cargo payload.
- Represents the permanent, logical communication between the original sender and final receiver.
- Travels end-to-end unchanged.

The Golden Rule: Every Ethernet frame carries exactly one IP packet. The frame contains the packet; the packet does not contain the frame.

Navigating the Globe: End-to-End vs. Hop-by-Hop



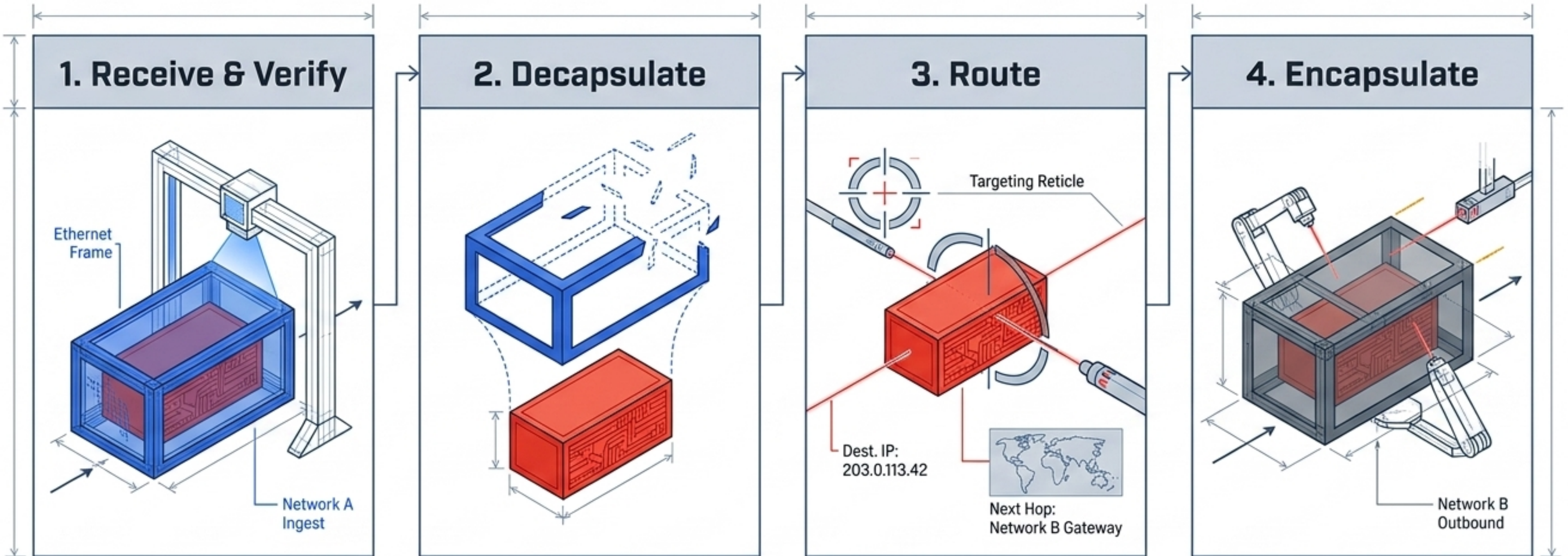
End-to-End (IP Addresses)

- The source IP (Paris) and destination IP (Tokyo) remain constant for the entire logical journey.
- Represents the true sender and receiver.

Hop-by-Hop (MAC Addresses)

- Ethernet frames **never travel** across the internet.
- They exist only between directly connected devices and are destroyed and rebuilt at every intersection.

What Happens Inside a Router?



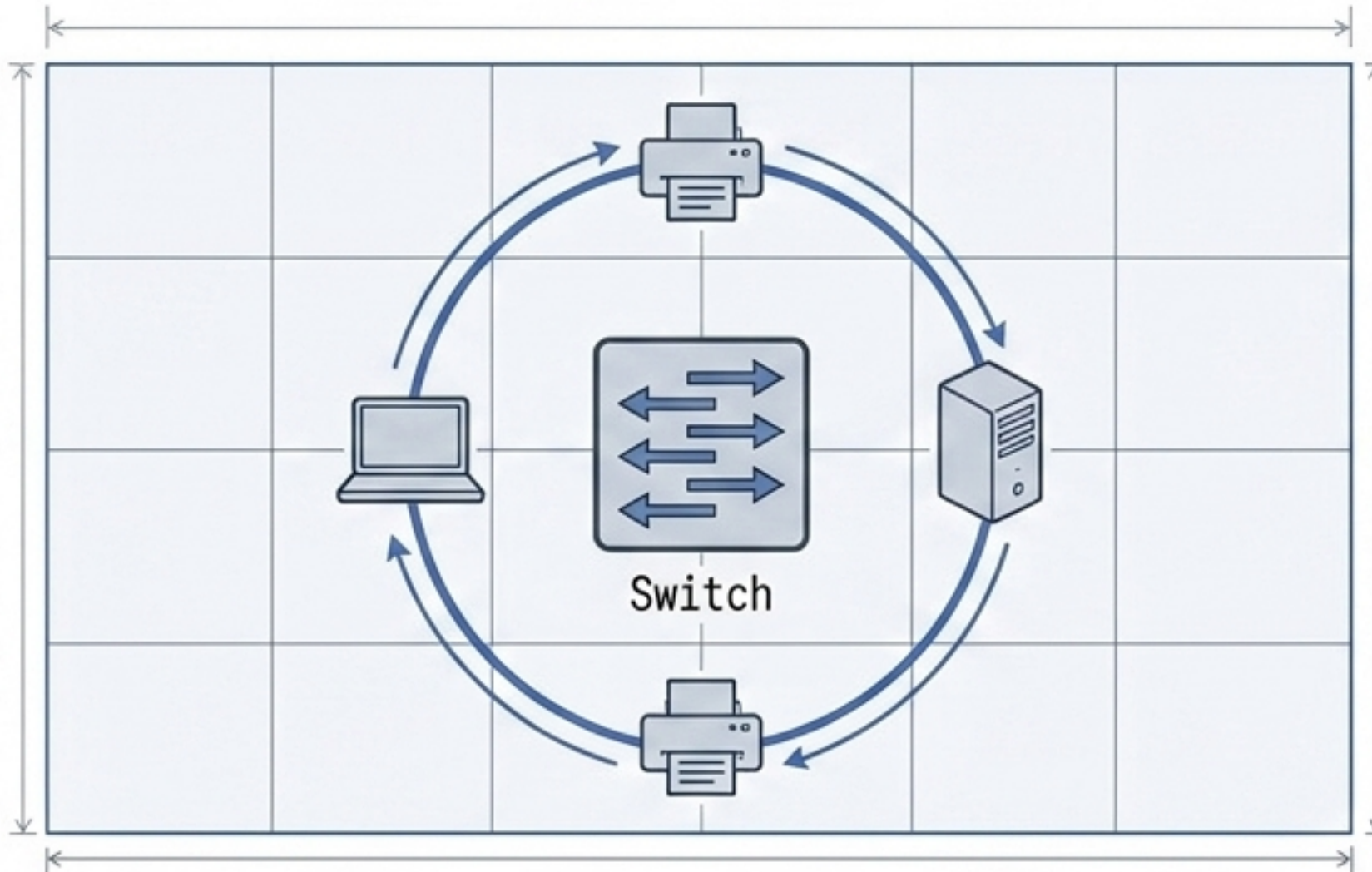
Router ingests the frame on Network A and verifies the Frame Check Sequence (FCS) for errors.

Router strips away the Link-layer header and trailer. The original Ethernet frame is destroyed.

Router reads the destination IP address on the exposed Packet to determine the next hop.
(Note: The router does NOT inspect application payloads like HTML).

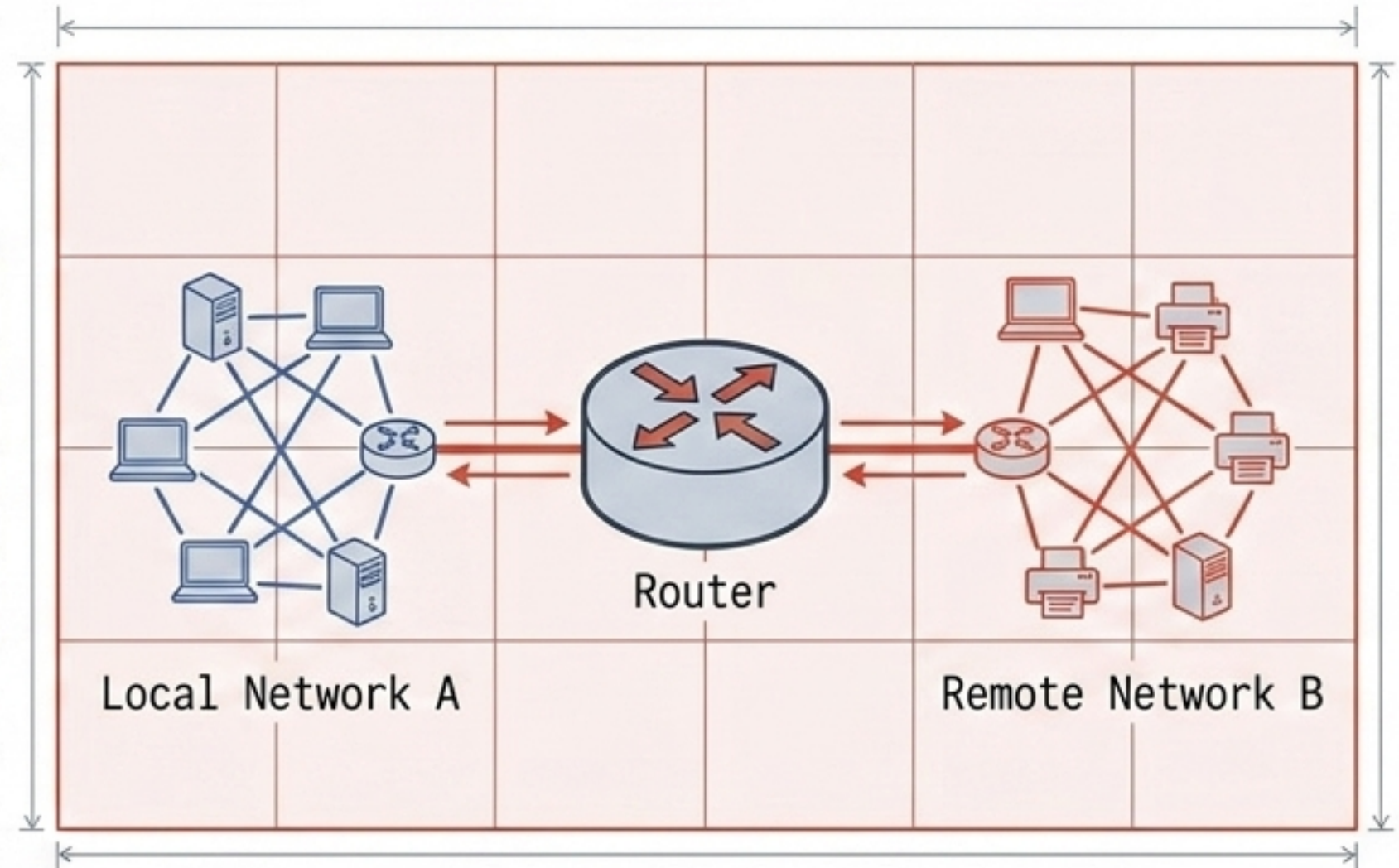
Router builds a completely new Ethernet frame addressed to the next specific device on Network B.

Forwarding Devices: Switches vs. Routers



The Switch

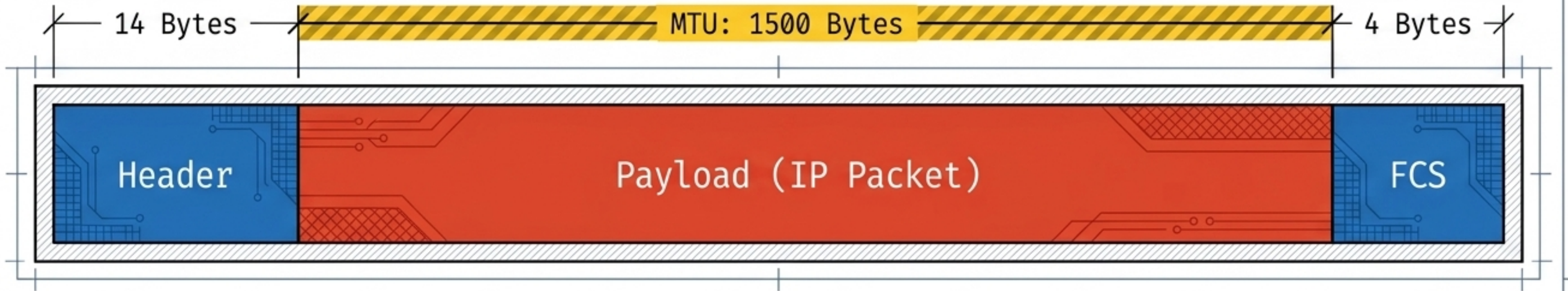
- Operates at Layer 2 (Link).
- Examines Destination MAC Addresses.
- Moves frames within a local, self-contained network.



The Router

- Operates at Layer 3 (Network).
- Examines Destination IP Addresses.
- Moves packets between different, remote networks.

Physical Limits: Maximum Transmission Unit (MTU)



The Concept

- Every network technology limits the size of a single frame to prevent monopolization of the medium, reduce latency, and manage memory buffers.

The Standard

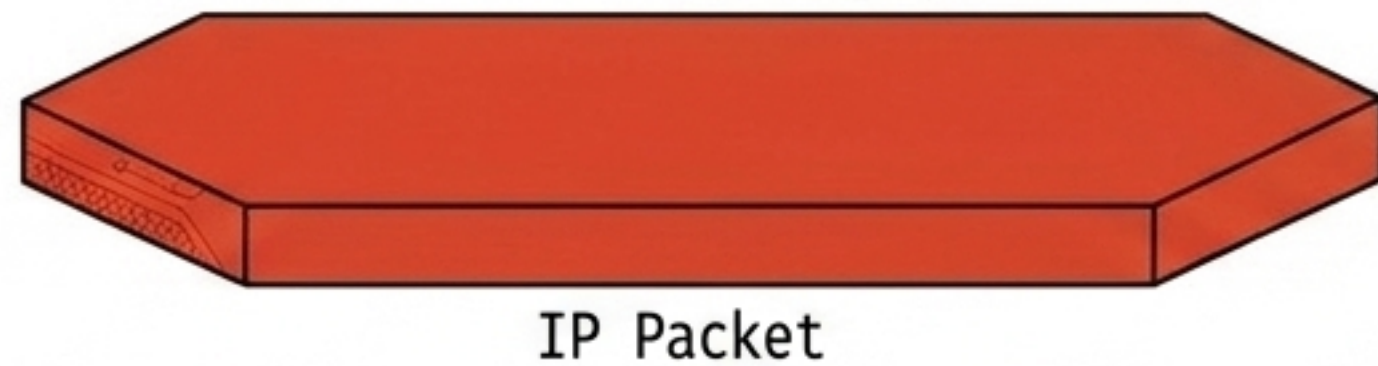
A standard Ethernet frame allows a maximum IP packet payload of exactly 1500 bytes. Data exceeding this must be handled specially.

Jumbo Frames

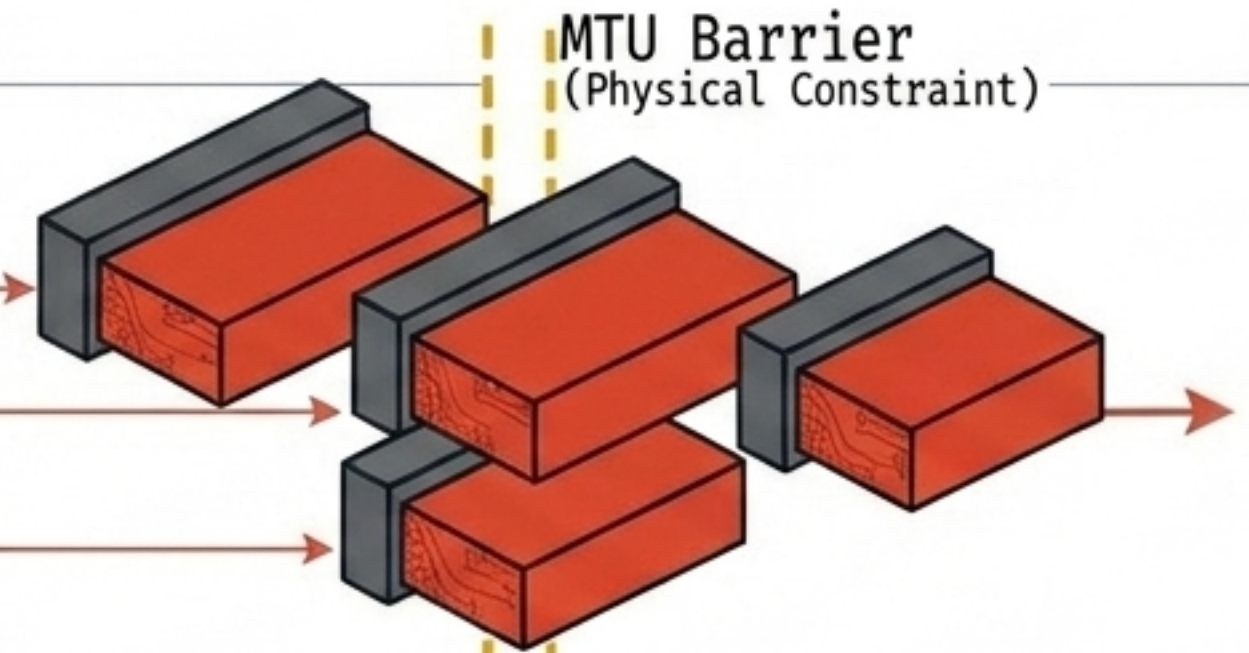
High-speed enterprise networks may support 9000-byte Jumbo Frames to reduce overhead, but every physical device along the path must explicitly support them.

Handling Oversized Packets

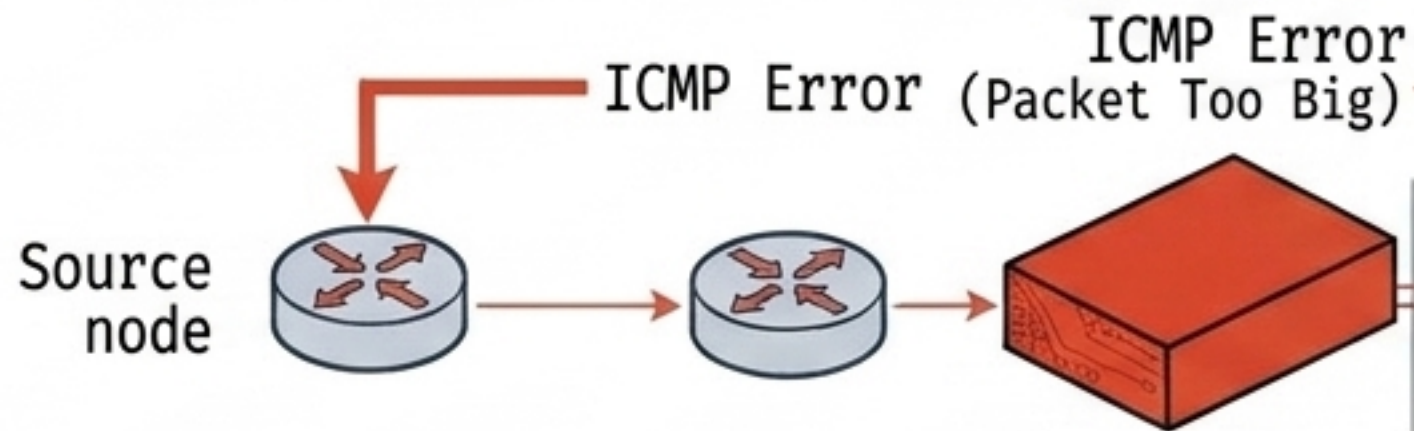
Legacy IPv4: Fragmentation



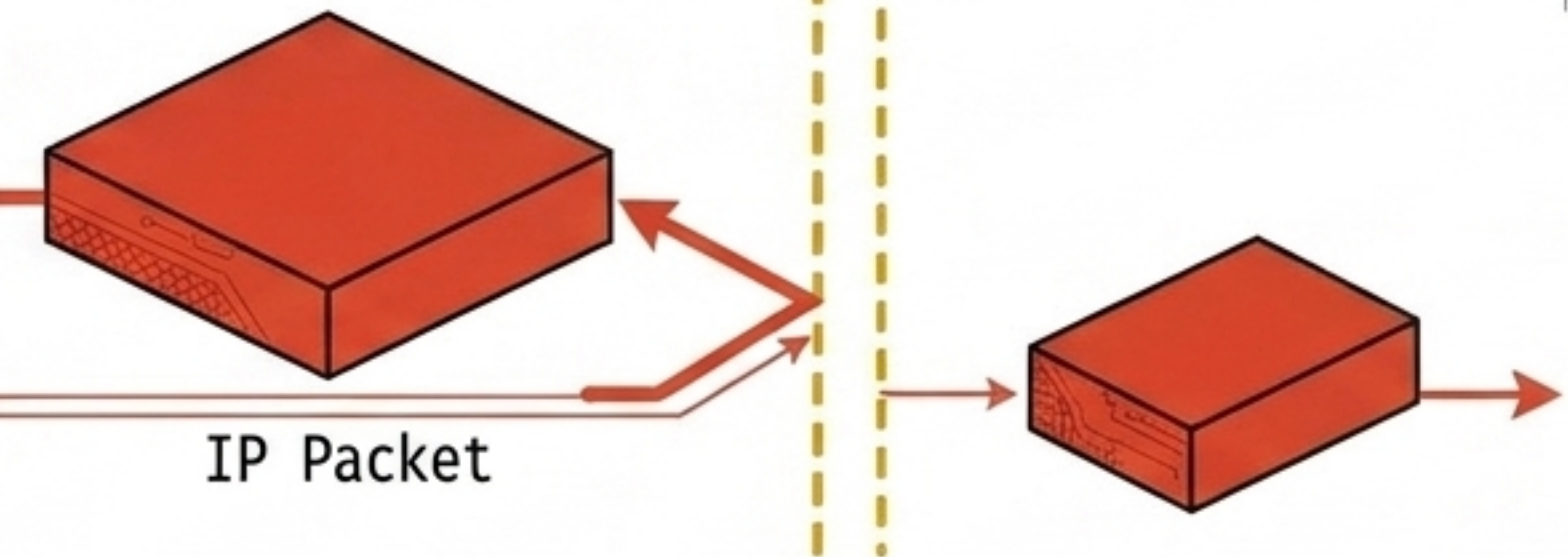
Routers divide oversized packets into fragments. Introduces heavy overhead, requires reassembly, and a single lost fragment ruins the entire transmission.



Modern IPv6: Path MTU Discovery (PMTUD)

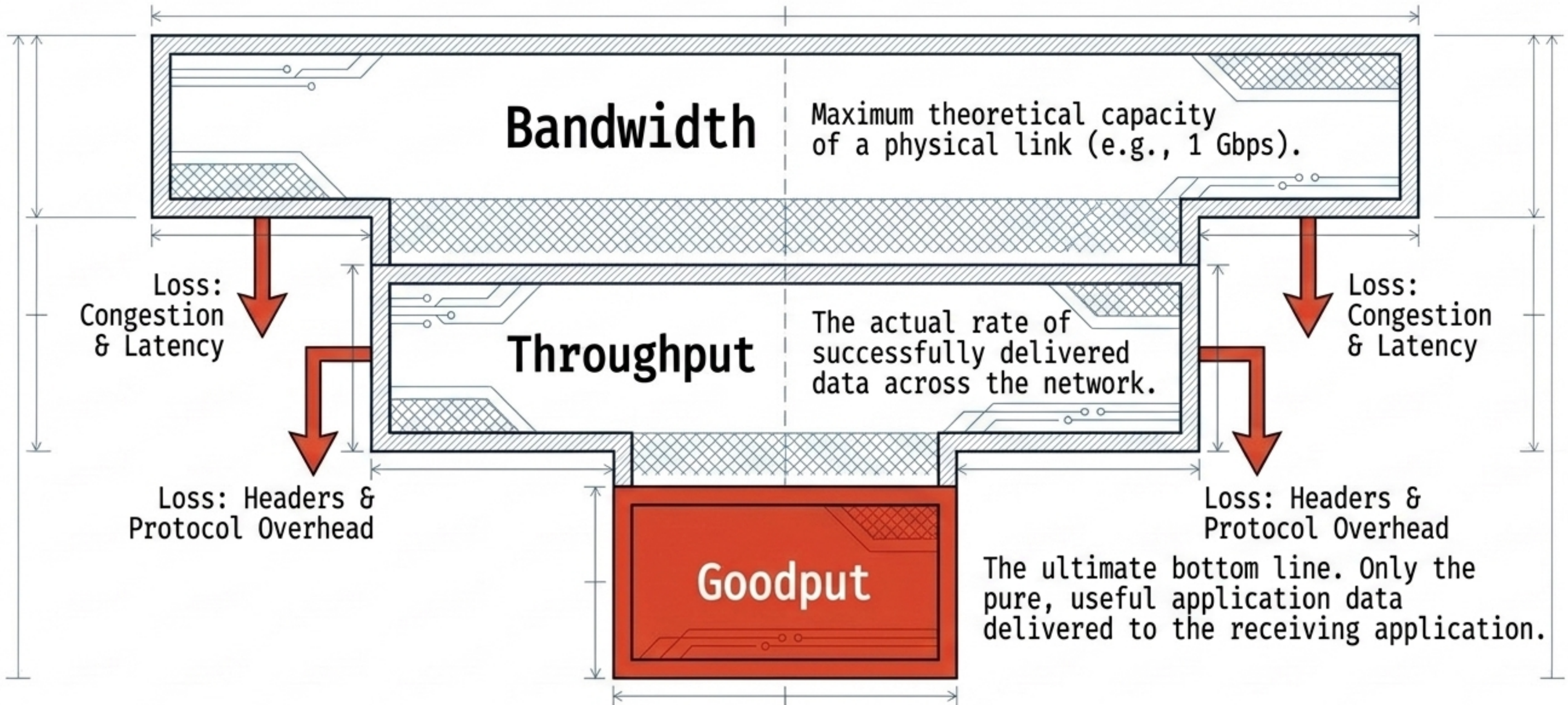


Routers drop oversized packets and send an error. sizes for dynamically eliminating fragmentation.

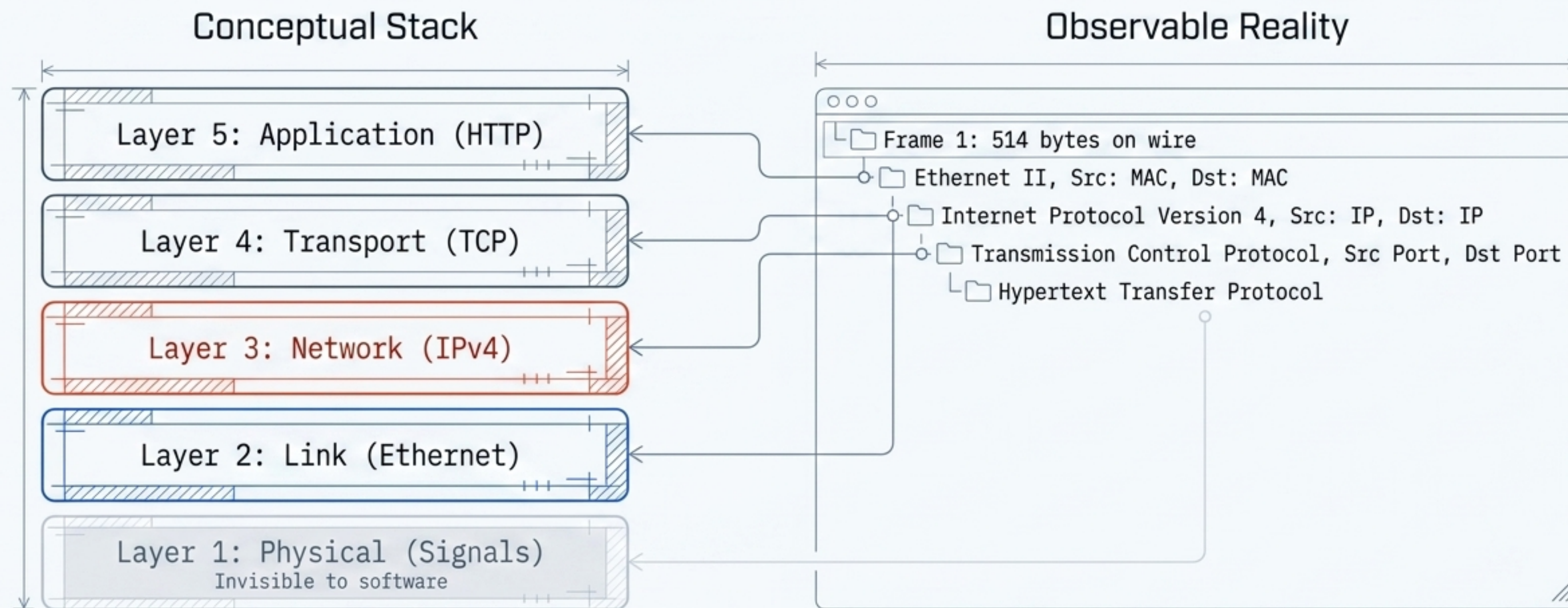


The sender dynamically discovers the largest usable for the entire path, eliminating fragmentation.

The Performance Funnel: Bandwidth, Throughput, Goodput



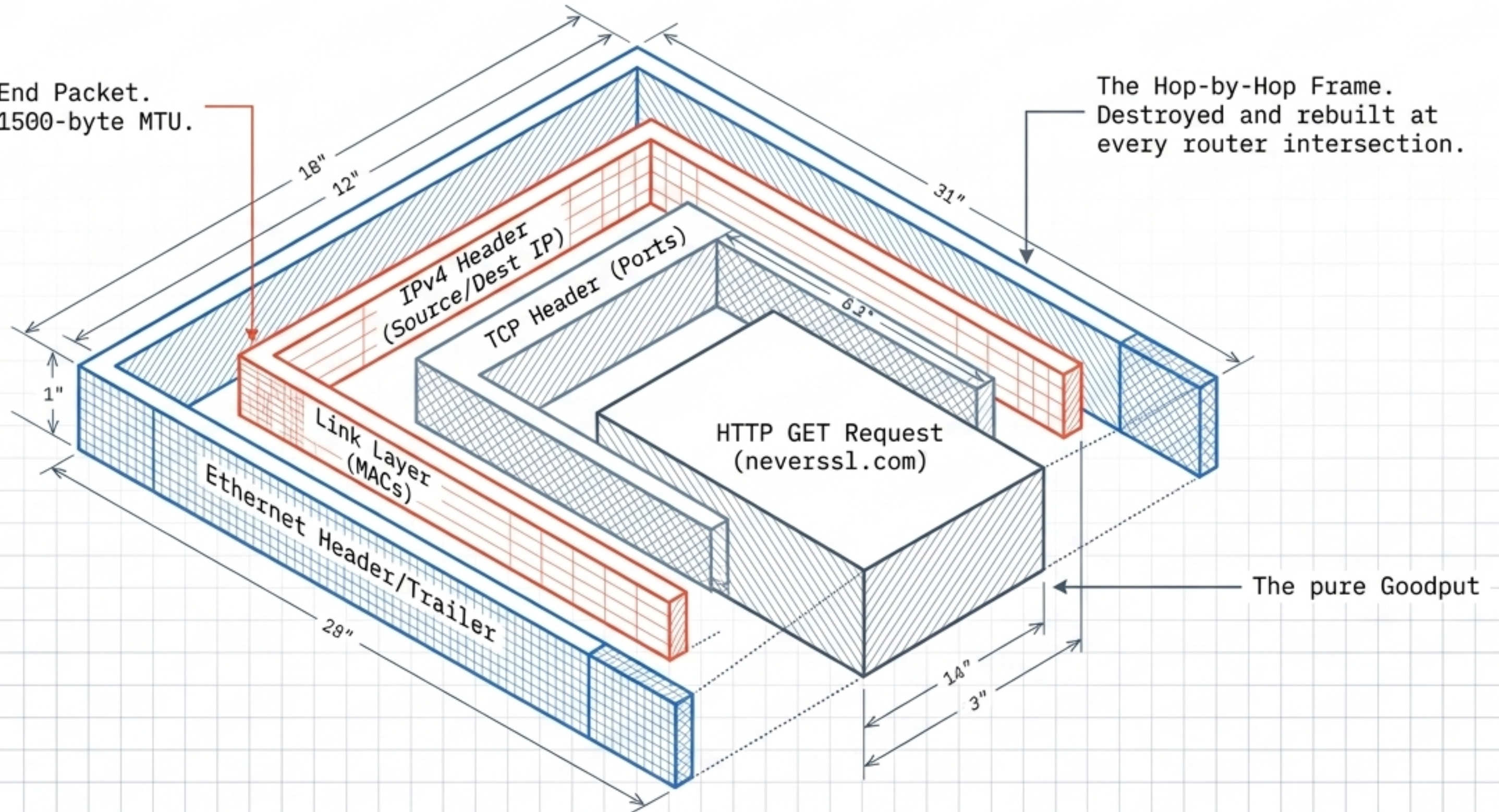
Observing the Protocol Stack in Wireshark



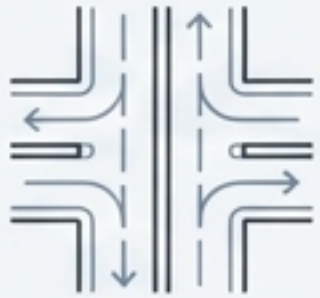
Wireshark captures traffic exactly as it arrives at your local network interface, translating abstract theories into observable, real-world data structures.

Synthesis: Deconstructing a Web Request

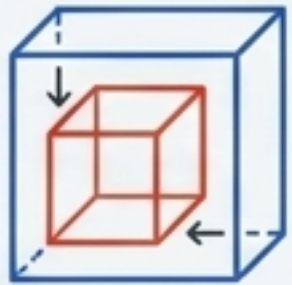
The End-to-End Packet.
Subject to 1500-byte MTU.



The Rules of the Road: Key Takeaways



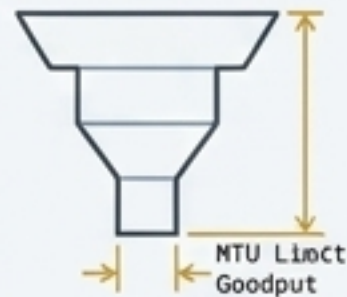
Architecture: The internet achieves massive scale via Packet Switching and statistical multiplexing, ensuring only active devices consume bandwidth.



Delivery: Packets (Network/IP Layer) travel end-to-end. Frames (Link/MAC Layer) travel hop-by-hop.



Routing: Routers strip, read, and replace Frames at every physical intersection, leaving the inner IP Packet untouched.



Limits: Data is strictly bound by physical size (MTU, typically 1500 bytes) and practical speed constraints, where true performance is measured by Goodput.